MSDS 451 Term Project Checkpoint A

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Introduction

The overall goal of this term project is to understand the quantitative methods utilized to create Exchange-Traded Funds (ETFs) in order to create an automated, algorithmic trading system capable of constructing and managing its own fund. An exchange-traded fund is a pooled investment security that operates similarly to a mutual fund that can be traded on the stock exchange like a regular stock. Historically, ETFs originally were defined strictly by market cap and sector, the first ETF being the one that tracks the top 500 companies in America (SPY). After the 2008 financial crisis, investors searched for algorithmic approaches. Much like Data Science itself, algorithmic ETFs started by defining and tracking a set number of factors, eventually leaning into machine learning techniques as AI grew in popularity.

The current ETF landscape is dominated by traditional, market cap-weighted funds like SPY and factor funds. While these funds are powerful in their own right, they largely assume stable market conditions. However, recently markets have been undergoing violent regime shifts due to external geopolitical factors. There is a significant gap for a low-cost, rule-based ETF that takes these drastic conditions into account without relying on human hedge fund managers.

Literature Review

The literature of financial machine learning is expansive and complicated, to say the least. This review will focus on the foundations of algorithmic ETFs, and the potential tools used in practice that could be applied in this term project.

The 1993 flagship paper by Fama and French established the fundamentals behind what is called factor investing. Fama and French demonstrated that a firm's size (market capitalization) and its value (measured by the book-to-market equity ratio) were significant, independent risk factors that explained the cross-section of average stock returns. From this, other factors were found that were able to accurately predict market volatility, such as momentum (trends over the past 6–12 months), quality (the firm's display of stable earnings and Return on Equity), and volatility.

This led to Smart Beta Investing, where multiple factors are weighted together to construct automated portfolios. For example, Monga et al. (2022) studied the integration of profitability, size, momentum, and low volatility factors to systematically select and weight stocks against a traditional cap-weighted index in the emerging India Equity market.
Another key area of focus for this analysis is investment strategies. Investment approaches can either be based on fundamental analysis, which predicts a security's future value using a variety of economic facts, or technical analysis, which focuses on market psychology and past price movements to forecast short-term trends. Harrington, C. (2003) provides a thorough comparison between the two philosophies.

Fundamental analysis is generally a long-term analysis, taking into account the company's general performance, financial statements, and recent company news. This process is long term, and "investors should be prepared to wait a considerable amount of time until a company's intrinsic worth is represented in the stock market." [1] Technical analysis, however, is more focused on the short-term implications of a stock price, utilizing time series data to make short-term predictions on market movements. Harrington, C. (2003) summarizes this point succinctly by suggesting that fundamental analysis works best as an investment strategy, while technical analysis works best as a trading strategy.

Another crucial aspect of this review is portfolio optimization. Modern Portfolio Theory, introduced by Harry Markowitz in 1952, established the framework for asset allocation by mathematically balancing expected return against portfolio variance. Due to the high sensitivity of this early model, practitioners developed the Black-Litterman model, which uses implied market equilibrium returns as a neutral starting point and mathematically blends them with the algorithm's specific market views to generate a stable, diversified portfolio. [2]

Lastly, when designing an automated ETF solution, risk management is a crucial piece of the puzzle. Value-at-Risk (VaR) is the standard financial metric used to quantify the maximum

[1] Harrington, C. (2003). "Fundamental vs. Technical Analysis." *CFA Institute*.
[2] Idzorek, T. (2006). "A Step-By-Step Guide to the Black-Litterman Model".

potential loss over a specific period using historical simulation. Quantitative funds increasingly rely on Expected Shortfall (ES), which calculates average expected loss in the tail end of the distribution, providing insight into severe market downturns. [3]

Methods

From the literature review, it is clear that assets, and decisions made on said assets, should be selected based on their market factors. For this term project, I have chosen the healthcare sector, as I am currently running a startup in healthtech and am curious on how healthcare will interact with booming AI trends, political regime changes, and environmental changes in the next 5 years. This limits the securities to US-listed equities within GICS Healthcare Sector, comprising of Pharmaceuticals, Biotechnology, Healthcare Equipment & Supplies, Healthcare Providers & Services, and Healthcare Technology. Asset selection will be determined based on company value, size, and quality. From there, buy and sell decisions will be based on stock momentum, quality, and technical analysis time series data.

The healthcare sector generally promotes long-term investments. While subsectors like biotech and healthtech can have high volatility, the healthcare sector is uniquely susceptible to massive, overnight price gaps that result from FDA approvals, clinical trial results, and acquisitions. A short squeeze on a biotech company that gets bought out could result in theoretically infinite risk, so risk management is significant in this industry. Due to this, I believe a fundamental investment philosophy based on economic factors, healthcare economic news feeds, and trailing 12-month price performance as its main contributors to asset decisions. Technical analysis will be used mainly in the healthtech and biotech sectors to find potential option opportunities.

One potential algorithmic approach that could find success in a sector like this is a quarterly rebalance of the ETF based on the quality, momentum, and consumer sentiment of the stocks. Potentially a small portion of the fund's assets could be used for options trading, though the majority of the ETF would remain in long-term bets. QMOM, the Quantitative Momentum ETF, has found success with this methodology. However, one key issue with this Quantitative momentum approach, and really all factor investment approaches, is its difficulty with regime shifts. As our goal was to create a regime shift stable ETF in the health sector, we can add volatility checks that can rotate capital out of equities and into safe haven assets such as bonds to preserve capital. For these regime-based portfolios, Hidden Markov Models have proved successful for tactical asset rotation [4], so that will be incorporated into our solution as well.

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[3] "Expected Shortfall – An Alternative Risk Measure to Value-at-Risk" (2020).

[4] *"Regime-Based Portfolio Allocation: A Hidden Markov Model Approach to Tactical Asset Rotation"* (2026).

Results

As the project so far is in the discovery phase, the results section will be utilized to discuss key insights from the literature review and observations surrounding the healthcare industry. The Healthcare sector is a sector that hosts a large variety of industries, from the stable pharmaceutical industry to the highly volatile biotech industry. Due to this, a healthcare ETF can benefit from long-term economic factor analysis, economic news sentiment analysis, while incorporating time series, momentum, and regime change data. Assets in this sector will be selected based on their quality, momentum, and volatility. The majority of the fund will focus on long-term bets which will be repositioned every 90 days, with around 10% of the fund positioned for high volatility options bets on the biotech and healthtech sectors. Finally, in order to combat macro environment regime changes, a Hidden Markov Model will allow for asset rotation when a regime shift is detected.

Discussion & Conclusion

As a person grounded in tech with little true finance knowledge, this literature review has been incredibly impactful. In my personal investments, I tend to focus on ETFs as they tend to be stable, growth-optimized funds for long-term investment. Regime shifts specifically impact my personal finances, especially in particularly volatile industries, so this activity has allowed me to develop a clear picture on why these regime shifts occur and how currently existing funds are able to combat the issue. Finally, the ability to programmatically build and rebuild an ETF is incredibly fascinating, from the mathematics behind the stock market to the architecture of the models used to build the trading fund.

There are still many open items between this conceptual design phase and the data and modeling phase, particularly the data pipeline and architecture of the solution. Next steps will include further research into quantitative finance companies and their solutions, not just the overall tools, but the architecture, timing, and validation. Along with this, exploratory data analysis will be key, especially in the regime shift section of the ETF automation.

Appendix: References

[1] Harrington, C. (2003). Fundamental vs. Technical Analysis: A Comparison of Philosophies.
[2] Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77–91.
[3] McNeil, A. J., Frey, R., & Embrechts, P. (2015). Quantitative Risk Management: Concepts, Techniques and Tools. Princeton University Press.
[4] Hamilton, J. D. (1989). A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. Econometrica, 57(2), 357–384.